

Element K

Summary of Design Process:

We reached out to the special needs classroom at Whitemarsh Elementary School because they wanted something to help with organization and engagement in the classroom. We chose to add LED lights to the classroom that could be useful for various types of activities, educational or fun. We consulted our stakeholders, Ms. Brennan and Dr. Sabatino, to confirm our idea. Then, we researched some lighting solutions that already exist and decided which aspects of them we wanted to incorporate into our design and which we wanted to manipulate. Next, we came up with a few important qualities for our design, such as safety, effectiveness, durability, etc. In Element C we formulated high priority and low priority requirements. The high priority being safety, aesthetic, durability, entertaining/effectiveness, and adaptability. The low priority requirements were easy to use, number of zones, and affordability.

For Element D we listed all the functions (providing light, control light, color, etc.) and then came up with 4 to 5 concepts for each and how we can achieve them. We also created a morphological chart and set up guidelines for what would be considered unacceptable, satisfactory, good, and excellent. After that, we came up with our design options where we put together the features we discussed and analyzed each of their benefits and drawbacks. Each design had different qualities. We talked about each criteria and came up with a weighting system and graded each of our designs to lead us to our final design choice (Rex). After getting feedback from our experts, we realized that we needed to do more research, which included looking into features of lights and how to measure them. We got 3 expert reviews (Mr Muscarella, Dr. Sarver, Dr. Dougherty). The feedback helped us look at our project in a practical sense, like looking more into the options we should have available for the lights. We were also given advice about time management and adjusting the brightness of the lights. We had to go back and review the aspects of our project to ensure that it was practical and intentional. We additionally restated our requirement in a chart and made sure they were viable, gave the evidence to prove it, and possible risks if they were not.

First we used an old project from a previous student to familiarize ourselves with how to work with LED lights and program them. We used Arduino to code some different patterns into the lights. Next we calculated how many lights we would need and ordered them. We used a table to set up a prototype, then used the app the lights came with and experimented with that. We documented this process with videos and screenshots of our work. We restated our high and low priority requirements and for each one we gave a detailed explanation of how we planned on testing them. We then conducted these tests and got data on how we could

improve our design. For Element J, we collected documentation of our conversations with our stakeholders

Lessons Learned:

- This project gave us experience with working with a client. We were able to just use our own ideas and had to reconvene with Mrs. Brenner and include her thoughts in our design too. Also the communication taught us how to be adaptable because we weren't always able to get in touch with Whitemarsh. We had to be able to continue our project without being restricted by this obstacle. We needed to learn how to not let our project come to a halt because of one roadblock and be able to keep moving on other parts of the project until the road gets cleared.
- This project taught us teamwork because every member has their own strengths and weaknesses. We had to be able to put members in their areas of expertise in order to work most efficiently. Also working in a team taught further taught communication skills because we had to be able to share ideas and settle disagreements.
- The project of LED lights strengthened our creativity. We had to take an already made invention and figure out how to make it our own. In our design process we had to look at previous LED light's designs and figure out our own changes, including games, shapes and installment patterns.

Value Judgement:

One aspect that went well in our project was that our group worked well together. We were able to get along without many disputes. Many members of the group have worked together in the past so we knew each other's strengths and were able to effectively split the work based on these strengths and weaknesses. Another thing that went well was that the lights we ordered turned out to be good quality and easy to use. (efficient/realistic)

Some areas that could have been improved are our focus. We got easily distracted which made the process go much slower. For example, on a shortened school schedule our group got very little work done because we were easily distracted by the idea of going home soon. Some of these occasions were caused by delays such as shipment or confusion on next steps. We also had some issues with the communication with our clients, leading to delays with planning when we could go in to meet our stakeholder(s). There were times where the responses were minimal so we had to find other ways to get in touch. Another area that we could improve on is our creativity. We had many ideas with the lights, but not all of them

were realistic. We could have worked more on thinking of more design choices that would be cheap, in our time restraint and still creative.